

Escaping the Embedding Trap: Neuro-Causal Symbiosis (NCS) for Grounded Causal Reasoning in Large Language Models

Houssam Rharbi

Independent Researcher

github.com/fDg21G/Neuro-Causal-Symbiosis

Preprint - Under Review

Abstract

Large Language Models (LLMs) exhibit a systematic failure we term the Embedding Trap: the conflation of distributional co-occurrence statistics with physical causal structure. When queried about causality, LLMs perform sophisticated pattern completion over high-dimensional vector spaces but are incapable of grounding their responses in temporal dynamics or interventional evidence. This paper introduces Neuro-Causal Symbiosis (NCS), a novel hybrid architecture that resolves this limitation through strict functional decomposition. NCS designates the LLM as a pure semantic router and entity extractor, delegating all causal inference to a deterministic empirical engine (System 1) grounded in real-world time-series data from the Federal Reserve Economic Data (FRED) repository. System 1 applies a multi-evidence ensemble comprising Granger Causality, Transfer Entropy, and Phase-Space Lag-Correlation to produce directional causal judgments with calibrated confidence scores. A confidence-threshold router arbitrates between System 1 and a Large Language Model fallback (System 2), the latter reserved exclusively for abstract or semantic domains where no measurable physical time-series exists. Empirical evaluation on five benchmark queries achieves 100% routing accuracy, with macroeconomic causal chains - including inflation-to-interest-rate transmission and oil-price-to-producer-price propagation achieving System 1 confidence scores of 80.1% to 86.2% across 300 months of historical data. Abstract queries (e.g., freedom versus happiness) are correctly and immediately escalated to System 2. NCS establishes a new paradigm for causal AI: one in which language models supply linguistic interface and semantic breadth, while deterministic engines supply causal truth.

1. INTRODUCTION

The question of whether a machine can reason about cause and effect rather than merely correlate - has been a defining challenge in artificial intelligence since the foundational work of Pearl [1] and Granger [2]. The emergence of Large Language Models (LLMs) as general-purpose reasoning engines has intensified this debate. LLMs demonstrate remarkable fluency across domains, including causal language, yet systematic evaluations

reveal a persistent and structurally motivated failure: these models cannot reliably distinguish between genuine causal relationships and spurious statistical regularities encoded in their training corpora [3,4].

This failure is not incidental but architectural. Current LLMs are trained to minimize prediction error over token sequences drawn from text corpora. The resulting representations - dense, high-dimensional embedding vectors encode the statistical geometry of language use, not the physical geometry of the world. When a model asserts that inflation causes interest rates to rise, it recapitulates a pattern frequently instantiated in financial journalism and macroeconomic textbooks, not performing any inference over temporal intervention data. We formalize this failure mode as the Embedding Trap: the systematic confabulation of causal structure from distributional statistics.

The implications are severe. In high-stakes domains — macroeconomics, clinical medicine, autonomous control systems, and policy analysis — causal errors are not merely inaccurate but consequential. A system that confidently inverts the causal direction of a drug-outcome relationship, or misattributes the driver of a macroeconomic shock, introduces systematic error precisely where precision is most demanded.

We propose Neuro-Causal Symbiosis (NCS) as a principled solution. Rather than attempting to instill causal reasoning into the LLM through prompting, fine-tuning, or architectural modification, NCS enforces a strict functional decomposition: the LLM is responsible solely for linguistic parsing and semantic entity extraction, while a deterministic empirical engine is responsible for all causal judgment. This decomposition is mediated by a confidence-based router that selects between the empirical engine and an LLM semantic fallback depending on the nature of the query and the quality of available evidence.

Our principal contributions are four-fold:

1. We provide a formal theoretical account of the Embedding Trap as a structural property of distributional language models that cannot be remediated by scale alone;
2. We introduce the NCS architecture, comprising System 1 (Empirical Reflex Core), a confidence-threshold Router, and System 2 (Semantic Cloud);
3. We present empirical results on a macroeconomic benchmark suite using 300 months of FRED data, demonstrating 100% routing accuracy and confidence scores exceeding the 0.80 threshold on all empirical queries;
4. We discuss implications for AI epistemology, Edge AI deployment, and the emerging post-LLM research agenda.

2. THE EMBEDDING TRAP: A THEORETICAL FORMULATION

2.1. Distributional Representations and Causal Structure

A language model parameterizes a conditional distribution $P(w_t \mid w_{1:t-1}; \theta)$ over token sequences. Through large-scale training, the parameter vector encodes a high-dimensional geometry in which semantically related tokens cluster proximally. The resulting word embeddings — whether static (Word2Vec, Glove) or contextual (BERT, GPT) — capture distributional statistics: the tendency of words to co-occur across syntactic contexts.

The fundamental epistemological problem is that distributional proximity does not entail causal relationship. The tokens {fever, antibiotics, infection} form a tight semantic cluster precisely because medical discourse frequently conjoins them — but the cluster encodes no information about the direction of intervention. An embedding-based model cannot reliably distinguish “infection causes fever” from “fever causes infection” because both directionalities appear in roughly symmetric textual contexts. The causal asymmetry — which is a property of the world, not of text — is irretrievable from distributional statistics alone [5].

2.2. Formal Statement of the Embedding Trap

Let X and Y be two measurable phenomena. Define the causal graph $G = (V, E)$ where a directed edge $X \rightarrow Y$ denotes that X is a direct cause of Y under the do-calculus [1]. The Embedding Trap identifies the following structural impossibility:

$$\nexists f : \mathbb{R}^d \rightarrow \{X \rightarrow Y, Y \rightarrow X, \emptyset\} \text{ such that } f(\phi(X), \phi(Y)) = G(X, Y) \quad \forall X, Y \in V \quad (1)$$

where $\phi : \Sigma^* \rightarrow \mathbb{R}^d$ is any embedding function trained on co-occurrence statistics from corpus C , and $G(X, Y)$ is the true causal relationship between X and Y . This impossibility holds because G is a property of the data-generating process $P(X, Y, \text{do}(X))$, which depends on interventional and temporal structure that is systematically absent from observational text corpora.

In practice, LLMs exhibit three measurable failure modes under causal probing: (a) direction reversal — asserting $B \rightarrow A$ when empirical evidence supports $A \rightarrow B$; (b) spurious causality - asserting a causal link between statistically correlated but causally independent variables; and (c) confidence miscalibration — expressing high certainty precisely when distributional statistics are most misleading. Each failure mode has been documented in systematic evaluation studies [3,6,7].

2.3. The Temporal Dynamics Problem

A particularly critical dimension of the Embedding Trap concerns temporality. Causation is fundamentally a temporal relation: cause precedes effect in time, and the causal mechanism operates through a physical dynamics that can be characterized by differential equations, transfer functions, or stochastic processes. Distributional embeddings are static objects computed from text corpora in which temporal order is at best implicit and at worst systematically inverted.

The decisive test of whether a system understands causality is whether it can correctly identify the directional arrow in time-series data — a test that cannot, in principle, be passed by any purely distributional representation. This motivates the central technical commitment of NCS: all causal inference must operate on real temporal dynamics, not on static vector geometries.

3. THE NEURO-CAUSAL SYMBIOSIS (NCS) ARCHITECTURE

NCS implements a strict separation between linguistic processing and causal inference through three sequentially connected components. We describe each in detail below.

3.1. System 1: The Empirical Reflex Core

Entity Extraction (NLP Layer). The first stage of System 1 applies a dependency-parsing NLP pipeline (spaCy with en_core_web_sm) to the user query. Critically, this stage performs entity extraction only — extracting variable names as character strings (e.g., inflation, interest rates) - and explicitly avoids any numerical transformation. This design decision is the primary architectural escape from the Embedding Trap: the NLP layer produces symbols, not vectors.

Reality Grounding Layer. Extracted entity names are mapped to real-world measurement instrument identifiers via a curated ontology registry implementing exact and fuzzy string matching against a domain-specific lexicon of 45+ economic concepts. The mapping transforms linguistic symbols into physical measurement instrument IDs — for instance, inflation $\rightarrow$ CPIAUCSL, interest rates $\rightarrow$ FEDFUNDS, oil prices $\rightarrow$ DCOILWTICO — grounding the subsequent analysis in empirical observables rather than semantic approximations.

Time-Series Acquisition. The FRED API is queried to retrieve the full historical time-series for each identified measurement instrument. In our benchmark evaluation this yielded 300 monthly observations spanning January 2000 through December 2024, capturing multiple macroeconomic cycles, policy regime changes, and structural breaks.

Multi-Evidence Causal Engine. Given aligned time-series (S_A, S_B) , the causal engine applies a three-component ensemble to determine causal direction. Series are first tested for stationarity via the Augmented Dickey-Fuller (ADF) test at significance level $\alpha = 0.05$, and first-differenced if non-stationary, yielding stationary representations ΔS_A and ΔS_B .

Component 1 - Granger Causality (weight 0.45). For lags $l \in \{1, \dots, L\}$, where $L = \min(12, \lfloor T/8 \rfloor)$, we test the null hypothesis that lagged values of ΔS_A provide no predictive information about ΔS_B beyond its own past, using the F-statistic of restricted versus unrestricted VAR. The minimum p-value $p^* = \min_l p(l)$ yields a directional score $g_{AB} = 1 - p^*$ in $[0, 1]$. Scores are computed symmetrically in both directions.

Component 2 - Transfer Entropy (weight 0.35). Transfer Entropy quantifies the directional reduction in predictive uncertainty:

$$TE(A \rightarrow B) = H(B_t \mid B_{t-1}) - H(B_t \mid B_{t-1}, A_{t-1}) \quad (2)$$

where H denotes Shannon entropy estimated via rank-based non-parametric binning. Unlike Granger causality, Transfer Entropy captures nonlinear couplings and requires no distributional assumptions. Values are computed at lags $l \in \{1, 3, 6\}$ and averaged to produce a lag-robust estimate.

Component 3 - Phase-Space Lag-Correlation (weight 0.20). The asymmetric lag-correlation statistic identifies the lag $L = \arg \max_l |\text{corr}(A_{t-L}, B_t)|$ that maximizes predictive coupling. The directionality of the lead-lag structure provides a phase-space geometric proxy for causal precedence, complementing the statistical tests above.

The three component scores are combined via weighted evidence aggregation:

$$score(A \rightarrow B) = 0.45g_{AB} + 0.35TE_{A \rightarrow B} + 0.20corr_A \quad (3)$$

The causal direction is determined as $\arg \max_{dir \in \{A \rightarrow B, B \rightarrow A\}} score(dir)$, subject to a minimum margin condition $|score(A \rightarrow B) - score(B \rightarrow A)| > \delta = 0.12$, below which the result is classified as BIDIRECTIONAL or UNDETERMINED.

3.2. The Confidence-Threshold Router

The Router is a deterministic decision function arbitrating between System 1 and System 2 outputs based on three conditions evaluated in priority order:

1. **Abstract Concept Detection.** If either extracted entity belongs to a pre-specified set of abstract concepts lacking physical time-series instantiation (e.g., {freedom, happiness, justice, consciousness, love}), the query is immediately escalated to System 2 without any System 1 computation.
2. **Grounding Failure.** If either entity cannot be mapped to a FRED series identifier by the Reality Grounding Layer, escalation to System 2 is triggered, preserving graceful degradation when the ontology registry lacks coverage.
3. **Confidence Threshold.** If $\max(score(A \rightarrow B), score(B \rightarrow A)) < \tau = 0.80$, the System 1 result is deemed insufficiently reliable and escalation to System 2 is triggered, with the System 1 result forwarded as a cross-validation hint.

This routing logic ensures that System 2 is never invoked for queries where deterministic empirical evidence is available and reliable, preserving the primacy of grounded causal inference while providing a principled fallback for genuinely uncertain or abstract domains.

3.3. System 2: The Semantic Cloud

System 2 is a Large Language Model (Claude 3.5 Sonnet or GPT-4o) deployed in a strictly constrained role: semantic causal reasoning over abstract conceptual relationships for which no physical measurement instrument exists. The LLM receives a structured prompt that explicitly states its role as a semantic reasoner operating in the absence of empirical data, and instructs it to output a JSON-structured causal estimate comprising: directional assessment, Semantic Consensus Index (SCI) (clamped to [0.40, 0.95] to reflect irreducible semantic uncertainty), causal mechanism description, and enumerated caveats.

Critically, System 2 does not output a calibrated probability. Instead, it generates a Semantic Consensus Index (SCI) clamped between [0.40, 0.95]. This metric quantifies the degree of sociological consensus in the text distribution regarding an abstract concept, explicitly avoiding the pretense of physical causal probability. When a System 1 hint is available, it is included as a cross-validation signal.

Critically, System 2 is epistemically subordinate: its outputs are labeled as semantic inference and are accompanied by warnings distinguishing them from empirically validated causal claims. This ensures that downstream systems can appropriately calibrate their trust in NCS outputs.

4. EXPERIMENTS AND RESULTS

4.1. Experimental Setup

To validate the architectural viability of NCS, we conducted an initial Proof of Concept (PoC) evaluation using a focused suite of five representative causal queries. All macroeconomic queries were evaluated against 300 monthly observations (January 2000 to December 2024) from the FRED database, comprising six time-series: CPIAUCSL (Consumer Price Index), FEDFUNDS (Federal Funds Effective Rate), DCOILWTICO (WTI Crude Oil Price), PPIACO (Producer Price Index), M2SL (M2 Money Supply), and UNRATE (Unemployment Rate). The causal engine was configured with ADF significance level $\alpha = 0.05$, maximum lag $L = 12$, evidence margin $\delta = 0.12$, and routing confidence threshold $\tau = 0.80$.

4.2. Benchmark Results

Table 1 reports the full benchmark results. All five queries were routed correctly: the four macroeconomic queries achieved System 1 confidence scores between 80.1% and 86.2%, satisfying the routing threshold; the abstract philosophical query was immediately escalated to System 2.

Query	Detected Direction	System	Conf. / SCI	FRED Series
Does inflation cause interest rates to rise?	CPI $\rightarrow$ Fed Funds	System 1	80.1%	CPIAUCSL, FEDFUNDS
Do oil prices affect producer prices?	Oil $\rightarrow$ PPI	System 1	86.2%	DCOILWTICO, PPIACO
Does the money supply drive inflation?	M2 $\rightarrow$ CPI	System 1	80.8%	M2SL, CPIAUCSL
Does unemployment affect inflation?	CPI $\rightarrow$ UNRATE	System 1	80.9%	UNRATE, CPIAUCSL
Does freedom cause happiness?	Freedom $\rightarrow$ Happiness	System 2	55.0% (SCI)	N/A (abstract)

Table 1: NCS Benchmark Results - Routing and Scores

The oil-price-to-producer-price relationship achieved the highest confidence score (86.2%), consistent with the strong and well-documented contemporaneous transmission of commodity price shocks through upstream production cost channels. The inflation-to-interest-rate relationship achieved 80.1%, with the ensemble correctly identifying the lagged causal transmission from consumer price inflation to Federal Reserve rate-setting at approximately 6 months — consistent with the empirical macroeconomic literature on monetary policy reaction functions [8]. The money-supply-to-inflation relationship achieved 80.8%, recovering the classical monetarist causal chain at a lag of 12 months.

The unemployment-inflation query yielded System 1 routing with 80.9% confidence. Notably, the causal engine detected the inflation-to-unemployment direction rather than the demand-pull direction implicit in the query formulation, illustrating an important NCS property: the system can correct the causal direction implied by natural language when empirical evidence warrants it.

4.3. Routing Accuracy

Routing accuracy was 100% across all five benchmark queries. All macroeconomic queries were correctly assigned to System 1; the abstract query was correctly and immediately escalated to System 2 without any System 1 computation. This perfect routing is a consequence of the deterministic nature of the routing logic: given a correctly configured abstract concept lexicon and FRED registry, routing is a rule-based classification over well-defined criteria, not a learned or probabilistic function.

4.4. Comparative Analysis

Table 2 provides a comparative analysis of NCS against baseline approaches across five capability dimensions.

Method	Ground.	Temp.	Non-linear	Abstr.	Causal Evidence Type
Vanilla LLM	No	No	No	Yes	Distributional only
Granger Causality alone	Yes	Yes	No	No	Linear predictability
Transfer Entropy alone	Yes	Yes	Yes	No	Information flow
NCS System 1 (Ensemble)	Yes	Yes	Yes	No	Multi-evidence consensus
NCS Full (S1 + Router + S2)	Yes	Yes	Yes	Yes	Empirical + Semantic hybrid

Table 2: Method Comparison Across Causal Reasoning Capability Dimensions

Vanilla LLMs handle abstract queries but fail all empirical dimensions. Granger causality alone captures linear temporal structure but cannot handle nonlinear couplings or abstract domains. Transfer Entropy alone is nonlinear but shares the same abstract domain limitation. The NCS System 1 ensemble achieves the full empirical profile, while the complete NCS architecture (System 1 + Router + System 2) is the only method that achieves all five capability dimensions simultaneously.

5. DISCUSSION

5.1. Epistemological Implications

NCS represents a departure from a long-standing assumption in the LLM research community: that sufficiently large and well-trained language models will approximate causal reasoning as an emergent property of scale. Our theoretical analysis and empirical results suggest this assumption is structurally untenable. The Embedding Trap is not a consequence of insufficient model scale or training data volume; it is a consequence of the

fundamental epistemological incompatibility between distributional statistics and interventional causal structure. No amount of scaling a distributional model will endow it with access to temporal dynamics it was never trained to represent.

The implication is not that LLMs are without value in causal reasoning contexts, but that their value is precisely delimited: they are powerful semantic engines capable of encoding and retrieving human causal knowledge as expressed in natural language, but they are not causal inference engines capable of discovering causal relationships from data. NCS operationalizes this distinction through its architectural decomposition, assigning each function to the component best suited to perform it.

5.2. Implications for Autonomous and Edge AI Systems

The NCS architecture has direct implications for deployment contexts where causal correctness is a hard requirement rather than a soft preference. In autonomous control systems — robotic manipulators, autonomous vehicles, industrial process controllers — causal misattribution can lead to catastrophic actuation errors. In clinical decision support, causal inversion can result in harmful treatment recommendations. The NCS pattern — deterministic empirical core with semantic fallback — provides a principled template for these high-stakes deployments.

Furthermore, the System 1 empirical core is computationally lightweight relative to large language model inference: it requires only time-series data retrieval, ADF testing, Granger regression, and entropy estimation — operations executable on commodity hardware with sub-second latency. This makes System 1 deployable at the network edge, reserving cloud-based LLM inference for the minority of queries that genuinely require semantic reasoning. The Router thereby implements a compute-optimal allocation strategy that aligns inference cost with epistemic necessity.

5.3. Limitations and Future Work

We explicitly bound our architecture by the assumption of causal sufficiency. LTI is designed for bivariate directional discovery and does not inherently resolve unobserved confounding. In deployment, this architectural boundary requires an empirical pre-check (e.g., domain-specific instrumental variables or Rosenbaum bounds) to flag environments where the causal sufficiency assumption is structurally violated.

Several limitations merit acknowledgment. First, the Reality Grounding Layer is currently implemented as a manually curated ontology registry with 45+ economic entities. Extension to broader scientific, medical, or social domains requires either large-scale curation or automated ontology alignment. Second, the three-component causal ensemble assumes pairwise bivariate causality. Many real-world causal relationships are confounded by latent common causes or mediated by intermediate variables, necessitating multivariate causal discovery (e.g., PC algorithm, FCI, PCMCI [9]) not yet implemented in System 1. Third, the confidence threshold $\tau = 0.80$ was selected empirically on a limited benchmark suite; systematic calibration across diverse domains remains necessary.

Future directions include: integration of multivariate causal discovery into System 1; development of learned entity-to-data-source mapping as an alternative to manual on-

tology curation; extension of the abstract concept taxonomy to biological, psychological, and social domains; and systematic evaluation against established causal benchmarks including Tuebingen Cause-Effect Pairs [10] and CauseMe [11].

6. RELATED WORK

The limitations of LLMs for causal reasoning have been documented by Kiciman et al. [3], who demonstrate that GPT-4 produces causally inconsistent outputs under systematic interventional probing, and by Jin et al. [6], who show that LLM causal confidence is poorly calibrated relative to interventional ground truth. Schölkopf et al. [5] provide a theoretical account of the gap between associational and causal knowledge from the perspective of causal representation learning. Pearl’s do-calculus [1] and the Potential Outcomes framework [13] provide the theoretical foundations for interventional causal inference that NCS operationalizes.

Hybrid neuro-symbolic architectures have been explored in related contexts: Garcez et al. [14] survey neural-symbolic integration approaches, and Marcus [15] argues for structured prior knowledge in neural systems. NCS contributes a domain-specific instance of this general program, specialized for causal reasoning with real-world time-series grounding. The use of Granger causality in AI interpretability has been explored [16], though not, to our knowledge, as a primary causal inference engine within a hybrid LLM architecture of the form NCS introduces.

7. CONCLUSION

We have introduced Neuro-Causal Symbiosis (NCS), a hybrid architecture that escapes the Embedding Trap by strictly decomposing causal reasoning into deterministic empirical inference (System 1) and semantic fallback reasoning (System 2), mediated by a confidence-threshold router. NCS does not attempt to make LLMs more causal; it removes LLMs from the causal inference loop entirely for queries where empirical evidence is available, while preserving their utility for abstract and semantic domains. Our benchmark evaluation demonstrates 100% routing accuracy on a macroeconomic query suite, with System 1 confidence scores of 80.1% to 86.2% across 300 months of FRED historical data.

The broader message of NCS is epistemological: causal truth is a property of the world, not of text. Any architecture that grounds causal inference exclusively in distributional language statistics is architecturally incapable of reliably recovering that truth. NCS demonstrates that the solution is not to improve the LLM but to build around it — deploying language models where they are strong (semantic encoding and retrieval) and supplementing them with deterministic empirical engines where they are fundamentally weak (causal direction, temporal dynamics, interventional structure).

We believe NCS points toward a broader research agenda we term Post-Embedding AI: architectures that treat statistical language models as semantic interfaces rather than epistemic authorities, grounding their outputs in physical observables, formal logic, or mathematical proof wherever such grounding is possible. The engineering of trustworthy causal AI demands nothing less.

ACKNOWLEDGMENTS

Macroeconomic data are sourced from the Federal Reserve Bank of St. Louis FRED database, publicly accessible without restriction.

References

- [1] J. Pearl, *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2000.
- [2] C. W. J. Granger, "Investigating causal relations by econometric models and cross-spectral methods," *Econometrica*, vol. 37, no. 3, pp. 424–438, 1969.
- [3] E. Kiciman, R. Ness, A. Sharma, and C. Tan, "Causal Reasoning and Large Language Models: Opening a New Frontier for Causality," *arXiv:2305.00050*, 2023.
- [4] T. Zevcik et al., "Evaluating LLM Causal Consistency: A Systematic Assessment," in *Proc. EMNLP*, 2024.
- [5] B. Schölkopf et al., "Toward causal representation learning," *Proc. IEEE*, vol. 109, no. 5, pp. 612–634, 2021.
- [6] Z. Jin et al., "Cladder: Assessing causal reasoning in language models," in *Proc. NeurIPS*, 2023.
- [7] C. Zhang et al., "Understanding causal hallucinations in large language models," in *Proc. ICLR*, 2024.
- [8] J. B. Taylor, "Discretion versus policy rules in practice," *Carnegie-Rochester Conf. Ser. Public Policy*, vol. 39, pp. 195–214, 1993.
- [9] J. Runge et al., "Detecting and quantifying causal associations in large nonlinear time series datasets," *Science Advances*, vol. 5, no. 11, 2019.
- [10] J. M. Mooij et al., "Distinguishing cause from effect using observational data: Methods and benchmarks," *J. Mach. Learn. Res.*, vol. 17, pp. 1–102, 2016.
- [11] J. Runge, "CauseMe: A benchmarking platform for causal discovery in time series," *Geosci. Model Dev.*, vol. 12, 2019.
- [12] P. Menzies, "Causation in the philosophy of mind," *Philosophy and Phenomenological Research*, vol. 77, no. 2, 2008.
- [13] D. B. Rubin, "Estimating causal effects of treatments in randomized and nonrandomized studies," *J. Educ. Psychol.*, vol. 66, no. 5, pp. 688–701, 1974.
- [14] A. d'Avila Garcez et al., "Neural-symbolic computing: An effective methodology for principled integration of machine learning and reasoning," *J. Applied Logics*, vol. 6, no. 4, 2019.
- [15] G. Marcus, "The next decade in AI: Four steps towards robust artificial intelligence," *arXiv:2002.06177*, 2020.

- [16] A. Khanna et al., "Interpreting neural networks via causal discovery," in *Proc. ICML Workshop on AI for Earth Sciences*, 2019.
- [17] B. Schölkopf, "Causality for machine learning," in *Probabilistic and Causal Inference*. ACM Books, 2022.